

Answer all question.

1. D, All of the mentioned
2. B, Minimum.
3. D, All of the mentioned.
4. B. regression
5. A, Decision tree.
6. A, Voting.
7. B., one
8. A, Bar chart.
9. C, Hyper planes.
10. B, Mean.
11. B, a real number.
12. C, Feature Selection.

- 13, B, regression.
- 14, C, time series.
- 15, B, logistic regression.
- 16, C, parameters of ML algorithm.
- 17, A, True
- 18, D, Non-diagonal element.
- 19, A, $TP / (TP + FN)$
- 20, ~~B, s.~~ C, Margin points.
- 21, Supervised learning is the machine learning that train model with labeled data, that the correct output are known, then to predict the new output, of dataset that didn't seen before. Example: linear regression, Decision tree, KNN, SVM.
- 28, Data pre-processing is the process of taking raw data, transform to format data, that can be understand and analyzed by computer.

29. the data normalization is required in KNN because it need to scaling data and it can represent within smaller range.
27. outlier is the data that abnormal from other structure. data, we can find out by use the library of code. which is box plot,
25. two major application of data science.
- Finance.
 - Healthcare.
31. K in K-mean algorithm is represent variable of number. that it can be express from 1 to 10.
23. Regression analysis is the process between two attribute. that can predict the another one if we know one.
22. the model ML are. good ~~star~~ output for accuracy since. the training ^{data} set are more accuracy than testing dataset.
24. two challenge of machine learning.
- the dataset are not well structure.
 - didn't know the specific goal.

26. Distinguish between the terms: structure and semi-structure data.
- Structured data: is the high level data, that is well prepared.
 - Semi-structure = is the low level data, that the data still have missing value, and not well prepared.
30. the continuous attribute are discretized because it is need to transform to set of data interval.
32. hyperparameters of machine learning algorithm.
- low level hyper parameter.
 - medium level hyper parameter.
 - high level hyper parameter.
33. gradient descent is the the data got error while processing.
35. it's set the data and preference the best data.
34. Comment on performance of classifiers.
- First the boundary are near the line so it's not a good performance.
 - Second the boundary are in the line so it's also not a good performance.
 - third the boundary are far from line so it's a good performance.